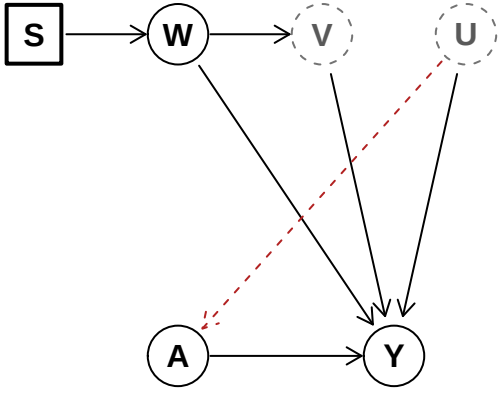
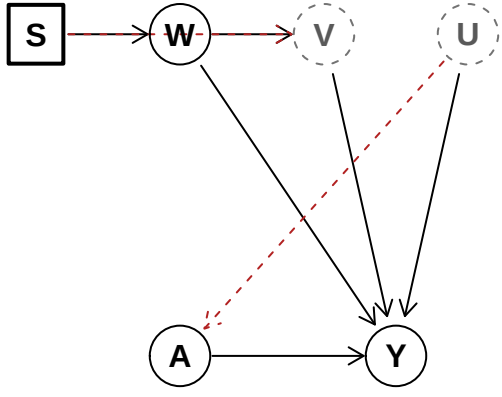
G_1: transportable



G_2: non−transportable